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The Editor
Technometrics

Dear Editor,

We submit the manuscript "Anytime-Valid Identification of Pareto-Optimal Alternatives from Noisy Simulations", by Raydonal Ospina, André Leite and Cristiano Ferraz, for consideration in *Technometrics*.

The manuscript addresses a question that arises whenever alternatives are compared by simulation on several performance measures: which alternatives can be declared Pareto-optimal, and which discarded, when the analyst decides how long to simulate after seeing the results. We show that the common rule of discarding an alternative when another is better up to a tolerance in every objective can discard the entire Pareto set, and we found this failure in our own software. The procedure we propose discards only on certified exact dominance, uses the tolerance only to accept, and summarizes the result by a certified cover; its statements hold simultaneously over time under any allocation and stopping rule. A simulation study and two applications with exactly known truth, to interval estimators for a binomial proportion and to inventory policies, evaluate it, and an R package implements it.

We believe the work fits the journal's interest in statistical methodology for simulation experiments and in methods accompanied by software. The main text has 25 pages; proofs, additional results and the code that reproduces every number are in the Supplementary Material. The manuscript has not been published and is not under consideration elsewhere. All authors approved the submission and declare no conflict of interest.

Sincerely,

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